Money and Crypto Portal

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Website: https://amrutha15102.github.io/Money-Portal/

Github: https://github.com/Amrutha15102

Purpose and implementation

This web application lets users explore historical currency rates, check cryptocurrency prices and calculate the current value of a crypto purchase. It uses separate HTML, CSS and JavaScript file. GitHub Pages hosts the website.

Frappe Charts 1.6.2 draws the charts. html2canvas 1.4.1 prepares PNG images. JavaScript fetch requests retrieve data, check responses and display loading or error messages. CSS Grid, Flexbox and media queries adapt the layout to smaller screens.

Data sources and calculations

Frankfurter: current and historical EUR exchange rates. Coinbase: current BTC, ETH and DOGE spot prices in USD. Alternative.me: the Bitcoin Fear & Greed Index, with source attribution. These are three separate API services.

The euro comparison combines Coinbase prices with the Frankfurter rate: coin value in EUR = coin price in USD / USD per EUR. Quotes and exchange rates can have different timestamps. Currency comparison charts use shared dates and index each series to 100 on its first shared date: index = 100 x rate / initial rate. This compares relative changes rather than raw currency units.

The purchase calculator uses the entered coin quantity and total USD cost, including buying fees. Current value equals quantity x current spot price. Percentage return equals 100 x (current value - cost) / cost. The purchase date records the transaction; the entered cost is used instead of reconstructing a historical purchase price. Selling fees are excluded.

User Manual

1. Choose start and end dates and a colour, then load historical rates. Click or tap a chart point for its date and value. The rate summary shows the minimum, maximum and first-to-last percentage change.
2. Load a crypto price, compare coins in euros, or load market sentiment.
3. Enter a purchase date, quantity and total cost, then calculate the current value. The classroom rule displays Sell at a return of at least +10%, Buy more at -10% or below, and Hold otherwise.
4. Add comparison charts according to your need. Load one currency or drag currencies onto a chart. Then add selected currency button provides a touch alternative. Set currency colours before adding or updating the comparison by clicking on the button beside chart color.(for example the default blue button in the first section)
5. Download SVG, or prepare a PNG and inspect its preview. Share image opens device sharing and download the PNG allows you to download the image.You can also downloas as csv.
6. In the what if I exchange my euro section,compare the usd received for a user-entered euro amount on the first and last available shared dates and the price difference.
7. After the above featured is used you can see the currency tracking below on the graph.

Assessment and verification

Criterion	Pts	Implementation / evidence
Drag and drop data	4	Currencies can be dropped onto comparison charts. USD, GBP, JPY can be dropped onto the charts in the comparison charts section.

Choose period	2	Start and end date controls are implemented for historical charts in all sections where comparison is required.
Choose colours	2	Single-series and per-currency colour controls are implemented. User can choose the color of the chart in every section for showing the trend or value in the charts by clicking on the color button beside chart color.
Add charts	2	User can add/remove independent charts in the my comparison charts section.
Click for details	2	Historical chart point shows date, rate and source when a point on the chart like the euro to usd chart is clicked.
Two API services	3	Frankfurter exchange rates and Coinbase crypto prices are implemented.
Third API service	2	Alternative.me sentiment API.
Merge and visualise	3	USD crypto prices converted to EUR with the exchange rate and charted.
Purchase calculator	5	Date, quantity and cost inputs generate current valuation, return and simulation rule.
Image download	2	User can export SVG and PNG images by clicking the save as svg/png buttons.
Social image sharing	2	Web Sharing through apps after loading the chart implemented.
PDF documentation	3	This report covers tools, calculations, usage and AI assistance.
Responsive interface	4	Responsive CSS and touch add button implemented. In every section user can access info, generate charts, pick colors, observe trends on the graph, see the highest and lowest values in a period. It is user-friendly.
Four browsers	3	Works in Safari, Chrome, Firefox and Edge.
Organised files	2	Separate index.html, style.css and app.js in the repository.

Additional features(with expected points):

Market summary dashboard.(3 points) : Displays BTC, ETH and DOGE prices and the Fear & Greed Index, with a refresh button, source attribution and an update timestamp.

Historical exchange scenario. (2 points): Compares the USD received for a user-entered euro amount on the first and last available shared dates. Shows dates above the converted amounts and reports the difference, excluding fees.

Currency movement ranking.(2 points) : Ranks USD, GBP and JPY by percentage change in units per euro over the selected period, using an ordered list and bar chart.

CSV data export. (2 points): Exports the loaded historical EUR–USD observations with dates, currency codes, rates and source attribution for spreadsheet analysis.

Declaration of AI usage

Github Copilot was used to assist with debugging code for the additional feature displaying the Fear and Greed Index and the visual glow effect applied to the subtitle. Chatgpt was used to generate the background image of notes in the title after drawing and giving the reference.